

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data from CSV to Delta Tables

Title: Load Customer and Order Data from CSV to Delta Tables

Description: The system must read customer and order data from CSV files and load it into delta tables, performing data cleansing and transformation as required.

Preconditions:

CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

Delta tables are available in the specified catalog and schema: `gen\_ai\_poc\_databrickscoe` and `sdlc\_wizard`

Main Flow / Functional Steps:

- Read customer data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a delta table `customer`.
- Read order data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a delta table `order`.
- Add a new column `TotalAmount` to the `order` table, calculated as the product of `PricePerUnit` and `Qty`.
- Remove null and duplicate records from both `customer` and `order` tables.

FR-DLT-002: Create and Load ordersummary Table

Title: Create and Load ordersummary Table

Description: The system must create an `ordersummary` table if it does not exist and load it with joined data from `customer` and `order` tables using SCD type 2 logic.

Preconditions:

- `customer` and `order` tables are available and populated with data.

`ordersummary` table does not exist or is available in the specified catalog and schema: `gen\_ai\_poc\_databrickscoe` and `sdlc\_wizard`

Main Flow / Functional Steps:

Create `ordersummary` table if it does not exist with the specified schema: `CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`.

- Join `customer` and `order` tables on `CustId` and load the joined data into `ordersummary` table using SCD type 2 logic.
- Maintain history in the `ordersummary` table by updating `StartDate` and `EndDate` columns accordingly.

FR-DLT-003: Update ordersummary Table on Customer Data Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the `ordersummary` table whenever there is a change in the `customer` table, using SCD type 2 logic.

Preconditions:

- `customer` table is updated with new data.
- `ordersummary` table is available and populated with data.

Main Flow / Functional Steps:

- Identify changes in the `customer` table.
- Update the `ordersummary` table by making old records inactive and new records active.
- Update `StartDate` and `EndDate` columns accordingly to maintain history.

FR-DLT-004: Create and Load customeraggregatespend Table

Title: Create and Load customeraggregatespend Table

Description: The system must create a `customeraggregatespend` table if it does not exist and load it with aggregated data from the `ordersummary` table.

Preconditions:

- `ordersummary` table is available and populated with data.

`customeraggregatespend` table does not exist or is available in the specified catalog and schema: `gen\_ai\_poc\_databrickscoe` and `sdlc\_wizard`

Main Flow / Functional Steps:

Create `customeraggregatespend` table if it does not exist with the specified schema: `Name`, `TotalAmount`, `Date`.

- Aggregate `TotalAmount` column from `ordersummary` table by `Name` and `Date`.
- Load the aggregated data into `customeraggregatespend` table.

FR-DLT-005: Implement using Delta Live Tables

Title: Implement using Delta Live Tables

Description: The system must implement the above functionality using Delta Live Tables.

Preconditions:

- Delta Live Tables are available and configured.

Main Flow / Functional Steps:

- Implement the data pipelines using Delta Live Tables to perform the required data transformations and loading.